# Graph

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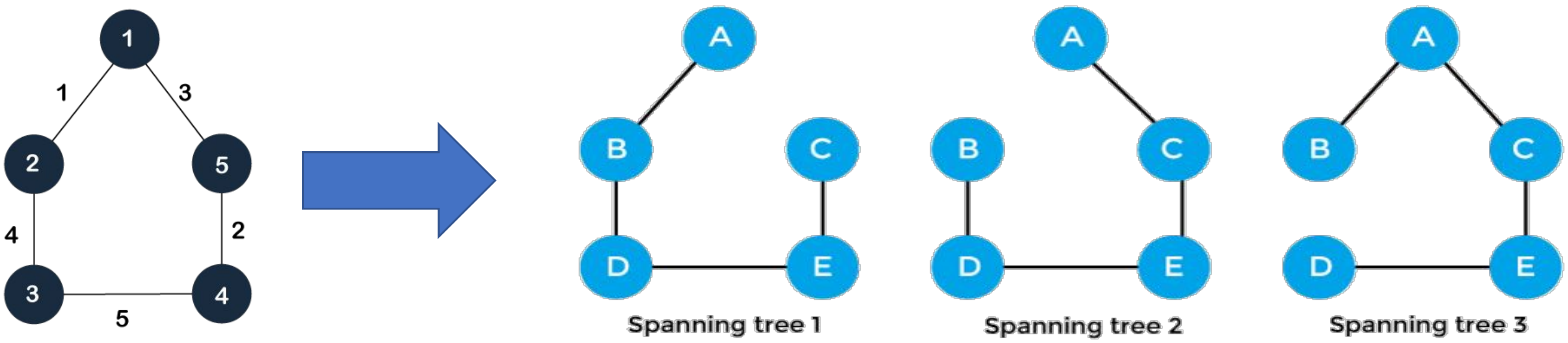
# Spanning tree

## 1. What are the properties of a spanning tree?

Here are some of the properties of a spanning tree:

1. It contains all the vertices of the original graph: Every vertex in the original graph is present in the spanning tree.
2. It is acyclic: A spanning tree does not contain any cycles.
3. It is connected: A spanning tree connects all the vertices of the original graph.
4. It has n-1 edges: If the original graph has n vertices, then the spanning tree has n-1 edges.
5. It is unique: A connected graph can have multiple spanning trees, but they all have the same number of edges.
6. Removing any edge disconnects the tree: If any edge in the spanning tree is removed, then the resulting subgraph will not be connected.
7. Adding any edge forms a cycle: If any edge is added to the spanning tree, then a cycle will be formed.



# Prim’s Algorithm (Greedy Method)

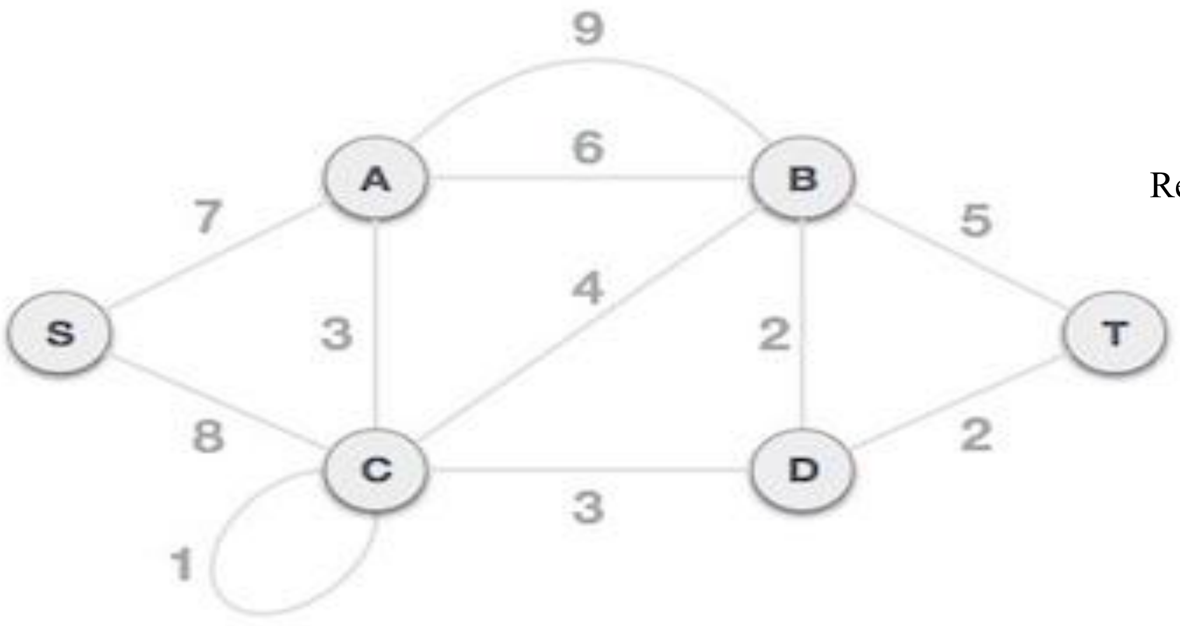
The minimum spanning tree from a graph is found using the following algorithms:

1. Prim's Algorithm
2. Kruskal's Algorithm

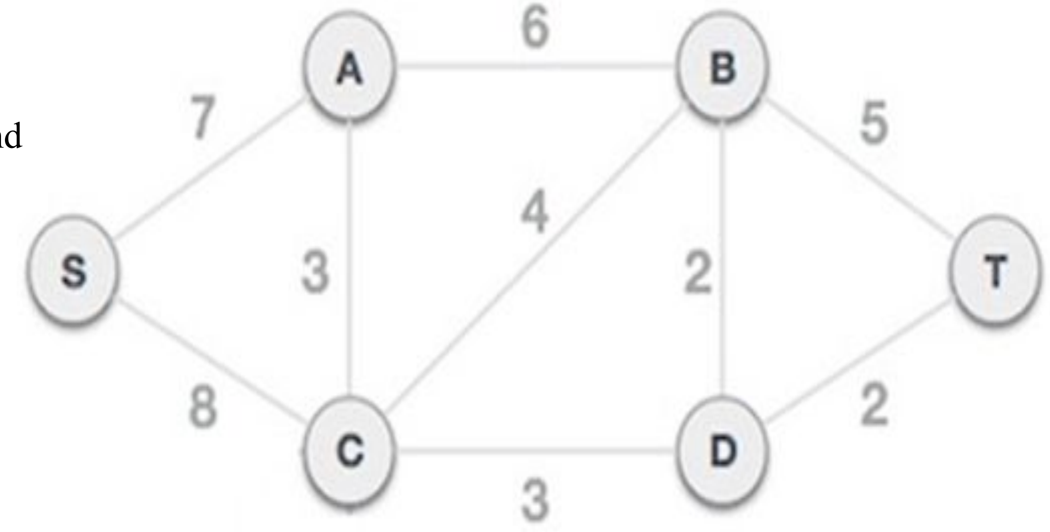
## Working procedure of Prim’s Algorithm

1. Remove all self-loops
2. Remove all parallel edges between two vertex except the one with least weight
3. Create a table where row and column will be exactly equal to the number of vertex.
4. Put the value of edges on the table. All diagonal elements will be 0. If no edge is exited between any two vertex, put ∞ .
5. Select a column as a starting node, and delete its row and indicate the node as finished.
6. Find the minimum weight from the column of the deleted nodes’ column.
7. Delete the row of the new node and indicate as the node as finished.
8. Repeat the process from 6-7 until finished.

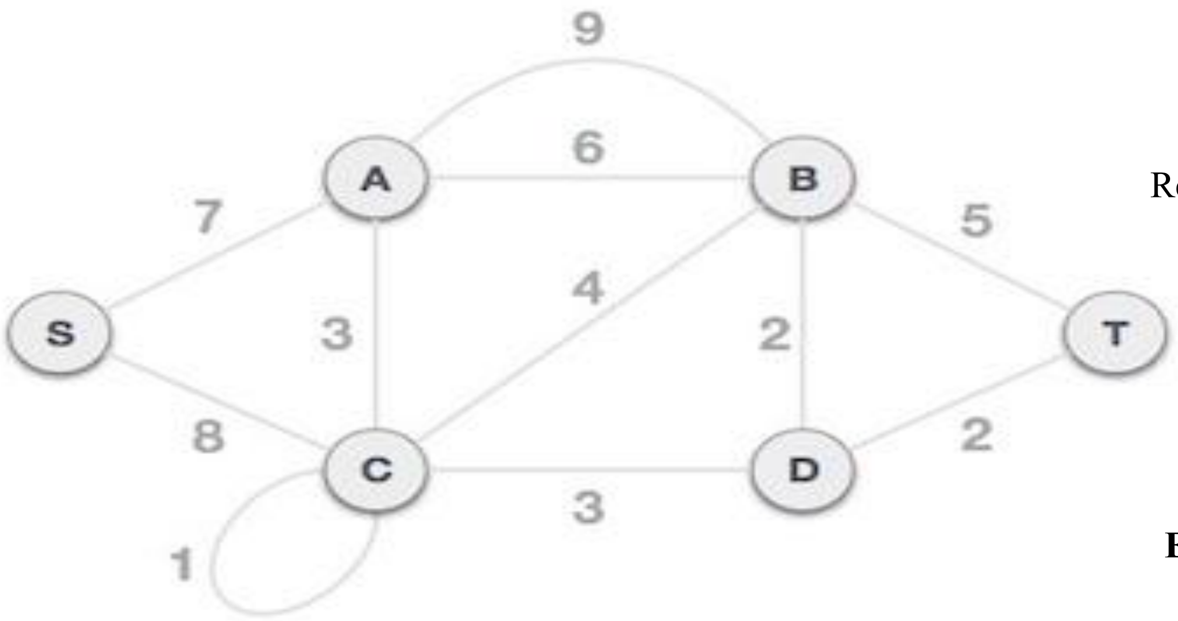
Find the minimum spanning tree from the following graph using Prim's algorithm.



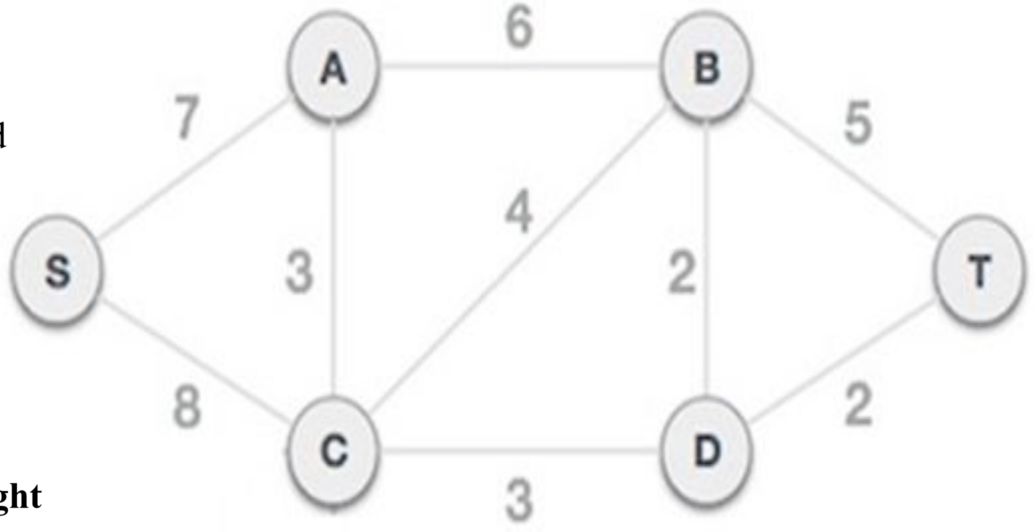

Remove all self and parallel edge.



Find the minimum spanning tree from the following graph using Kruskal algorithm.




Remove all self and parallel edge.




| **Edge** | **Weight** |
|---|---|
| DT | 2 |
| DB | 2 |
| CD | 3 |
| CA | 3 |
| CB | 4 |
| BT | 5 |
| AB | 6 |
| SA | 7 |
| SC | 8 |

Sort all the edges in ascending order based on weight.

# Thank You